

Lab 3: Architectural Justification

Problem Statement #53 — Freelance Content Creator Escrow Platform

We chose the Microservices Architecture for the Freelance Content Creator Escrow Platform System.

Architectural Choice

The system is decomposed into independently deployable services: a Media Upload & Watermarking Service, a Review & Approval Service, a Milestone Escrow Service, an Auth & Identity Service, a Notification Service, and a Persistence/Database Service. Services communicate through well-defined provided/required interfaces (DraftMedia, ApprovalDecision, PaymentStatus, UserIdentity), matching the Microservices style.

Two Scenario-Related Reasons

- 1. Independent scaling for media processing:** The NFR requires automated watermarking to complete in under 5 seconds for video/image drafts, a compute-heavy task. Isolating this in its own Media Upload & Watermarking Service lets it be scaled or given dedicated (e.g. GPU-backed) infrastructure without over-provisioning the rest of the platform.
- 2. Fault isolation between media and money:** Creator uploads and reviews happen far more frequently than payment releases, and are far less critical to get perfectly right on every request. Separating the Milestone Escrow Service means a spike or bug in media handling cannot crash or degrade the payment-release path, and vice versa.

Security Advantage

Keeping the Milestone Escrow Service as its own microservice, behind its own required UserIdentity interface and reachable only via the ApprovalDecision interface from the Review & Approval Service, narrows the attack surface around funds. Sensitive payment logic and credentials never need to sit inside the same process or codebase as public-facing media upload endpoints, so a compromise of the upload service does not directly expose payment-release capability.

Performance Benefit

Because each service is independently deployable, the Media Upload & Watermarking Service can be horizontally scaled during high-traffic periods (e.g. many creators uploading drafts at once) without needing to scale the Escrow or Notification services, keeping the watermarking latency requirement met while controlling infrastructure cost.